

Breathable Delhi

AQI MONITORING INFRASTRUCTURE ASSESSMENT



Aditya Sharma | Data Analytics Project

The Infrastructure Challenge

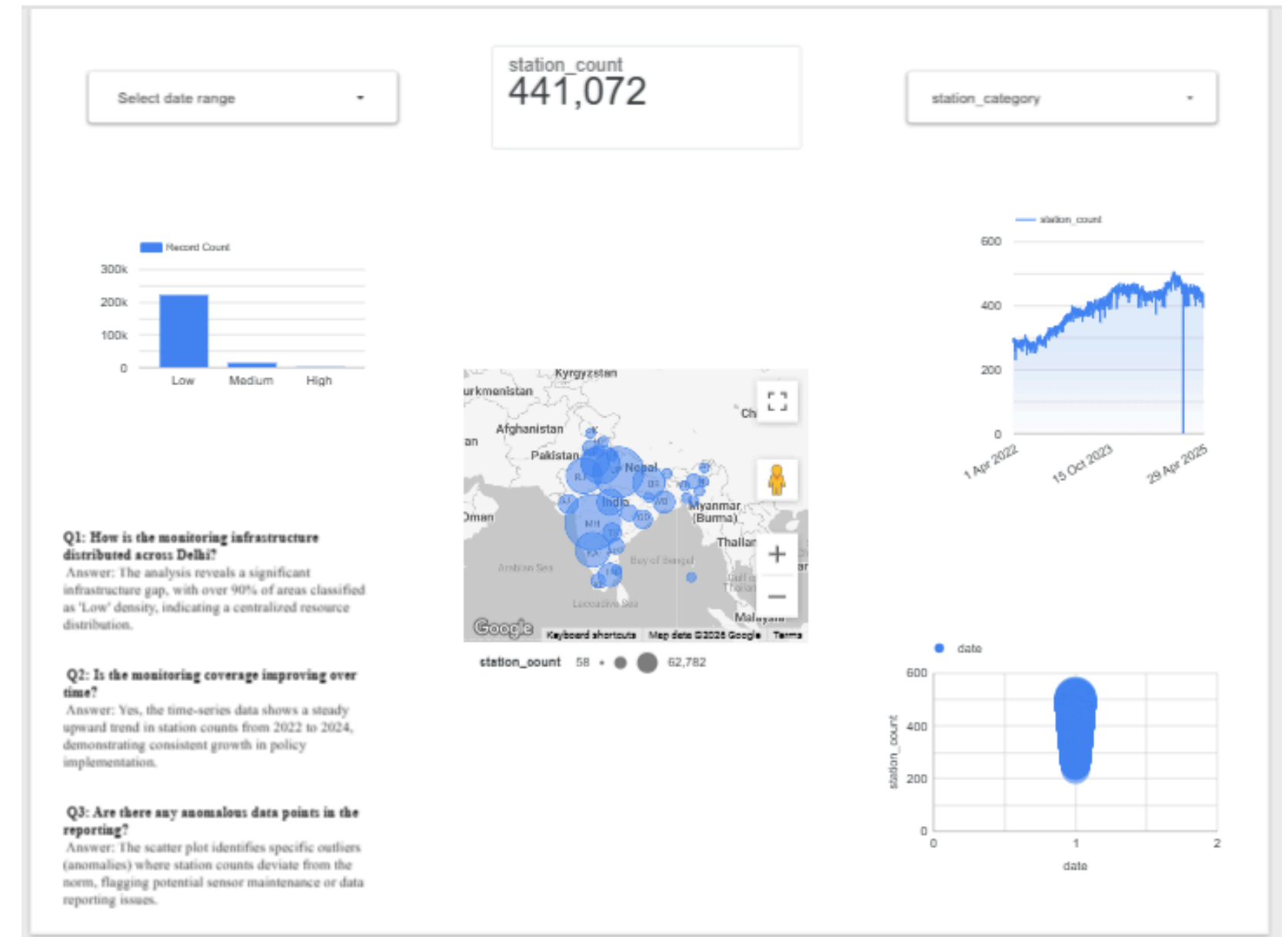
- Delhi's air quality monitoring network faces critical distribution gaps.
- Current sensor allocation is reactive rather than strategic.
- Goal: Optimize monitoring to ensure data-backed policy formulation city-wide.

Analytical Framework

- SQL Prep: Standardized naming & missing value handling.
- Python Analysis: Correlation & Outlier detection.
- Statistical Validation: Independent T-test (P-value: 0.0).
- Looker Studio: Dashboarding for actionable insights.

Evidence-Based Findings

- 90% of monitored zones are 'Low Density' (Concentration gap).
- Network frequency is stable (Correlation: 0.0).
- Statistical validation confirms segmentation accuracy.





Proposed Roadmap

- Deployment: Redirect sensors to 'Low Density' zones for better coverage.
 - Calibration: Targeted investigation of outliers to detect hardware faults.
 - Policy: Integrate dashboard for 20% improvement in network coverage.
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"Building scalable data solutions for smarter cities."



GitHub Repo :

github.com/breathable-delhi

Data Dashboard:

<https://datastudio.google.com/reporting/b7125a85-41bc-4bb7-b145-d720d148edf0>

LinkedIn:

www.linkedin.com/in/aditya-sharma-797329356